PSIT Kanpur | PRANVEER SINGH INSTITUTE OF TECHNOLOGY

KANPUR

5G FIVE GLITCHBREAKERS

TECH EXPO 2026

STUDENT PROJECT EXHIBITION

PROJECT NAME

FLOORGUARD

IoT-Based Anti-Theft Floor Mat System

Footstep Detection | Capture Image | Buzzer Alert | Mobile Notification

Team 5G – Five Glitchbreakers

Team Members

1. Adarsh Srivastava (Leader)
2. Ankit Yadav
3. Kushagra Awasthi
4. Abhishek Pandey
5. Arpit Chaturvedi

02

PROBLEM STATEMENT

When homes are unattended, security gaps can become costly.

- Traditional locks cannot notify owners when an intruder gets inside.
- CCTV may record an incident, but someone must notice and respond.
- Theft can be discovered only after valuables are already missing.

The gap: real-time detection and immediate alerts at the point of entry.







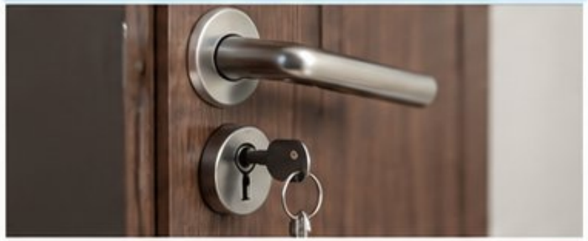


Real-world examples of intrusion, theft and delayed awareness.

03 EXISTING SOLUTIONS AND THEIR LIMITATIONS

Several home security solutions are available in the market, but they have certain limitations.

Traditional Door Locks



Limitations:

- No alert or notification
- Can be easily bypassed
- Cannot detect unauthorized entry through windows or other areas
- No real-time visibility

CCTV Cameras



Limitations:

- Only records footage
- Requires manual monitoring
- No instant alert to the owner
- Does not actively prevent or deter intruders
- May miss blind spots

Smart Home Sensors (PIR, Door/Window Sensors)



Limitations:

- Detects only at specific points (door/window)
- Can be avoided by intruders
- No information about who entered
- Limited coverage inside the house
- Separate installation required at each entry point

Smart Video Doorbells



Limitations:

- Works only at the main door
- Does not detect entry from other areas inside the house
- Requires internet and power
- Does not trigger an alarm/buzzer
- Higher cost for full home coverage

Gap: Existing solutions either only record, require manual monitoring, have limited coverage, or do not provide immediate alerts at the point of entry.

04

OUR PROPOSED SOLUTION

FLOORGUARD – IoT-Based Anti-Theft Floor Mat System

1. Intrusion Detection
When someone steps on the floor mat, the pressure sensor detects the movement.

2. Capture Image
The camera module captures a photo of the person immediately.

3. Alert Mechanism
Buzzer sounds at the location to alert nearby people.

4. Mobile Notification
A notification is sent to the owner's phone with the captured image.

5. Live Monitoring
The owner can view live footage through the mobile app to see who has entered.

System Working Diagram / Prototype Image

05 HARDWARE COMPONENTS AND TECHNOLOGIES USED

FloorGuard is built using affordable and readily available hardware components integrated with IoT technology.

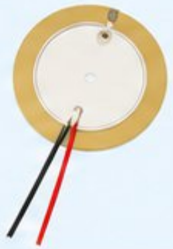
01



Arduino UNO R3

Main controller to process sensor data and trigger actions.

02



Piezoelectric Sensor (35mm)

Detects footsteps on the floor mat (using vibration/pressure).

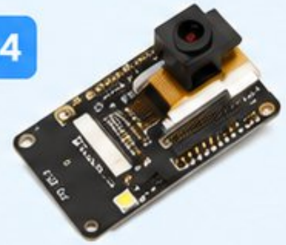
03



ESP8266 NodeMCU

Provides Wi-Fi connectivity to send alerts and images.

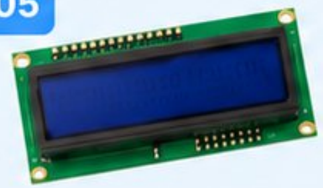
04



ESP32-CAM Module

Captures and sends live images of the person at the entrance.

05



16x2 LCD Display

Displays system status and alerts.

06



Active Buzzer (5V)

Sounds an alarm when intrusion is detected.

07



Micro Servo Motor (SG90)

Can be used for a mechanical lock/ indicator (optional).

08



Power Hub

Distributes required power to all modules (5V/3.3V).

09



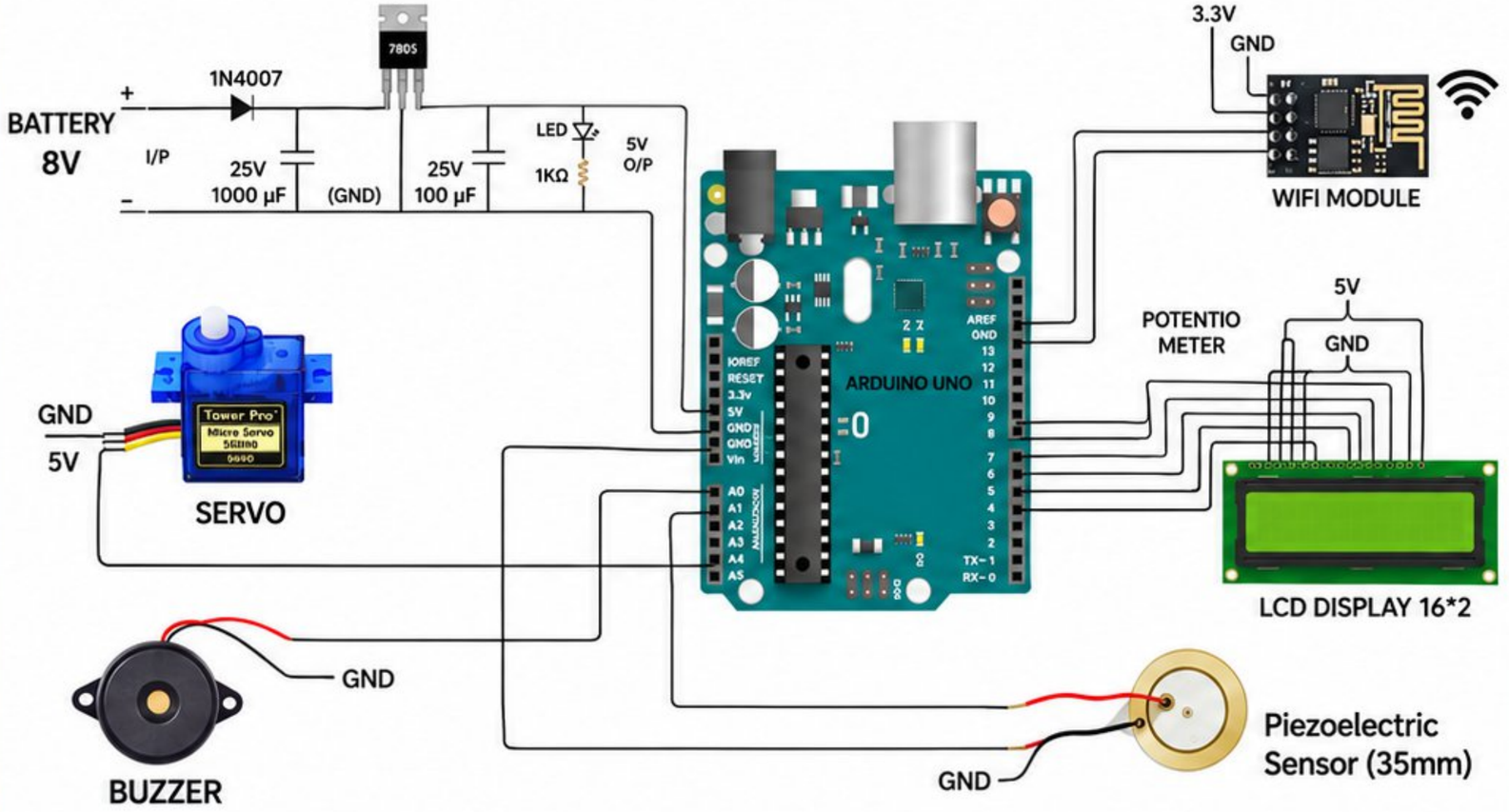
Voltage Converter

Steps down voltage to provide stable 5V supply to components.

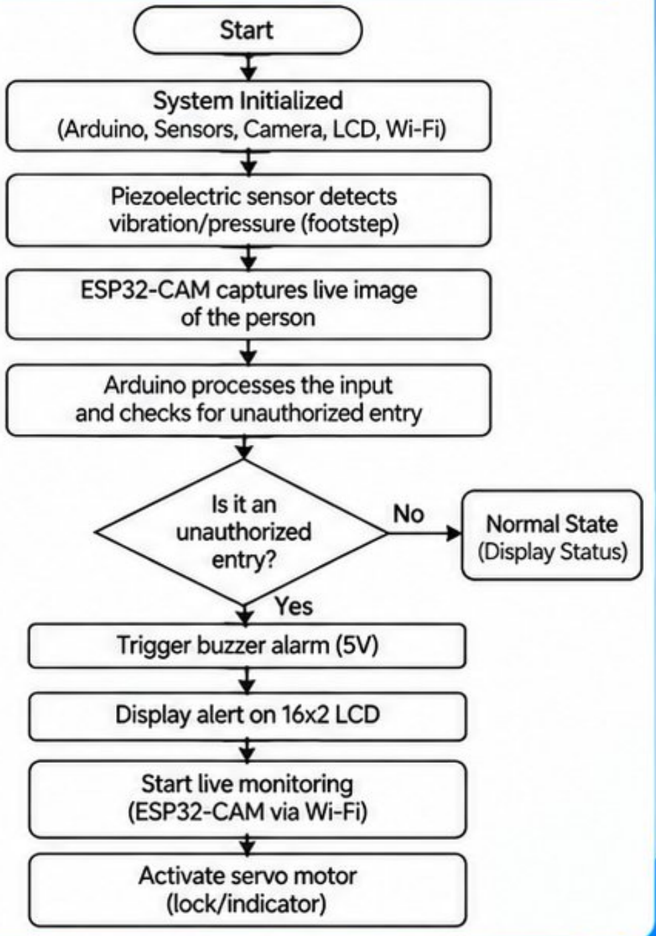
06 SYSTEM ARCHITECTURE AND WORKING WORKFLOW

Hardware circuit diagram and step-by-step working flow of FloorGuard.

System Architecture (Circuit Diagram)

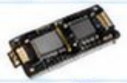


Working Workflow (High Level Flow)



07 COST ESTIMATION OF THE SOLUTION

FloorGuard is a low-cost and easily deployable solution built using readily available components.

S.No.	Component	Specification	Image	Cost (₹)
1	**Arduino UNO R3**	Main controller		₹ 550
2	**Piezoelectric Sensor**	35mm (vibration/pressure)		₹ 30
3	**ESP8266 NodeMCU**	Wi-Fi module		₹ 250
4	**ESP32-CAM Module**	Camera module		₹ 450
5	**16x2 LCD Display**	Display for status & alerts		₹ 180
6	**Active Buzzer**	5V buzzer		₹ 30
7	**Micro Servo Motor**	SG90 (for lock/indicator)		₹ 160
8	**Push Button Switch**	User input (optional)		₹ 10
9	**Perf Board + Jumper Wires**	For connections		₹ 100
10	**5V Power Supply**	Adapter/USB (regulator)		₹ 150
11	**Connecting Wires, Misc.**	Wires, connectors, etc.		₹ 70

Total Cost of Our System



₹ 2,380/-

Key Highlights

- **Low-Cost & Affordable**
  Total cost ~ ₹2,380/- only.
- **Readily Available Components**
  All components are easily available in the market.
- **Easy to Build & Install**
  Simple hardware setup with minimal wiring.
- **Scalable & Customizable**
  Can be enhanced with additional features as per requirement.

PSIT Kanpur

PRANVEER SINGH INSTITUTE OF TECHNOLOGY
KANPUR

5G
FIVE GLITCHBREAKERS

08 IMPACT, BENEFITS AND PRACTICAL APPLICATIONS

FloorGuard provides an affordable, smart and effective solution to enhance physical security using IoT.

Impact

Prevents Theft and Unauthorized Entry
Detects intruders at the entrance in real-time, reducing theft cases.

Improves Safety & Security
Provides an additional layer of security for homes, offices and sensitive areas.

Immediate Real-time Alerts
Instant buzzer, LCD alert and live camera monitoring for quick action.

Affordable Modern Security
A low-cost alternative to expensive CCTV and alarm systems.

Peace of Mind
Helps users feel safer about their belongings, family and workspace.

Benefits

Low Cost
Total cost ~ ₹2,380/- only, making it affordable for everyone.

Easy to Install & Use
Simple hardware setup with minimal wiring and maintenance.

Real-time Detection & Alerts
Immediate buzzer, LCD notification and live camera feed via Wi-Fi.

Live Monitoring
ESP32-CAM provides real-time images of the person at the entrance.

Scalable & Customizable
Can be enhanced with mobile app, cloud storage or multiple mats.

Energy Efficient
Low power 5V system, suitable for continuous operation.

Practical Applications

Homes
Protects houses, apartments and villas from intruders and theft.

Shops & Retail Stores
Monitors entry at shops, showrooms and commercial spaces.

Offices & Workspaces
Secures office entrances and restricts unauthorized access.

Warehouses & Storage Areas
Useful for godowns, storages and restricted zones.

Labs & Research Facilities
Protects valuable equipment and confidential areas.

Restricted/High-Security Areas
Can be deployed in sensitive zones like server rooms, labs, archives, etc.

Overall Impact

FloorGuard makes security smarter, simpler and more accessible by combining IoT, sensors and real-time monitoring at a very low cost, helping to create safer homes, workplaces and public spaces.

09 CONCLUSION AND REFERENCES

Conclusion

FloorGuard detects pressure/footstep activity on a protected mat using a piezoelectric sensor.

Arduino UNO processes the sensor input and activates local alerts (buzzer, LCD, servo, etc.).

ESP32-CAM and Wi-Fi support visual monitoring and remote awareness with real-time images.

Final Takeaway

An affordable additional layer of security — not a replacement for locks or CCTV systems.

Research References

01 **Samalla, K., Chandaka, L., Ushaswini, L., Likhitha, M., & Jahnavi, M. (2024).**
"Design and Implementation of an IoT-Based Anti-Theft Mat Using Arduino Nano, ESP32 Camera, and Piezo Sensors."
International Journal for Research in Applied Science and Engineering Technology (IJRASET), 12(6), 61933.
DOI: 10.22214/ijraset.2024.61933 | https://doi.org/10.22214/ijraset.2024.61933

02 **Hidayah, Q. H., Laksana, R. P., Prabowo, A., & Simatupang, D. S. (2026).**
"IoT-Based Home Security System: ESP32-CAM Integration and Real-Time Notification."
Indonesian Journal of Electrical Engineering and Electronics (ITEE), 4(2), 149–161.
DOI: 10.33050/itee.v4i2.956 | https://doi.org/10.33050/itee.v4i2.956

03 **Selim, K. K., Yehia, H. M., & Saleeb, D. A. (2024).**
"Energy Harvesting Floor Tile Using Piezoelectric Patches for Low-Power Applications."
Journal of Vibration Engineering & Technologies, 12, 8613–8622.
DOI: 10.1007/s42417-024-01379-z | https://doi.org/10.1007/s42417-024-01379-z

04 **East Japan Railway Company (2008).**
"Demonstration Experiment of the 'Power-Generating Floor' at Tokyo Station."
JR East Group Press Release, January 11, 2008.
URL: https://www.jreast.co.jp/e/development/press/20080111.pdf

Japan Example – Piezoelectric Flooring

JR East (Japan) tested piezoelectric flooring at Tokyo Station ticket gates in 2008 to convert footsteps into electricity. (Source: JR East Press Release, 2008)

PSIT
Kanpur

PRANVEER SINGH INSTITUTE OF TECHNOLOGY
KANPUR

5G
FIVE GLITCHBREAKERS

Thank You!

We appreciate your time, attention and valuable feedback.

Safer Spaces | Smarter Solutions | Innovative Ideas | For a Better Tomorrow





FLOOR GUARD

An IoT Based Anti-Theft Floor Mat System

DETECT | ALERT | MONITOR | PROTECT